AI

(Intro to Artificial Intelligence)



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AI Content Generation

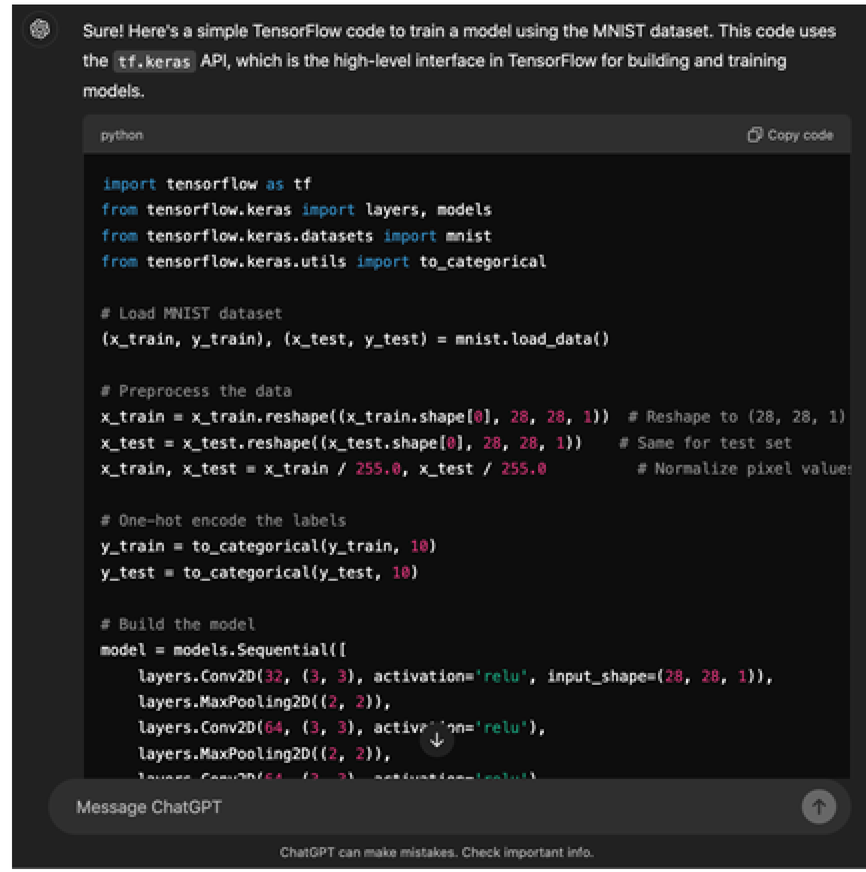
Generating images/video from natural language:

"A snowy view of Old Port in Montreal"

AI Content Generation

Generating programs from natural language:

> "write code in tensorflow to train a model using MNIST"



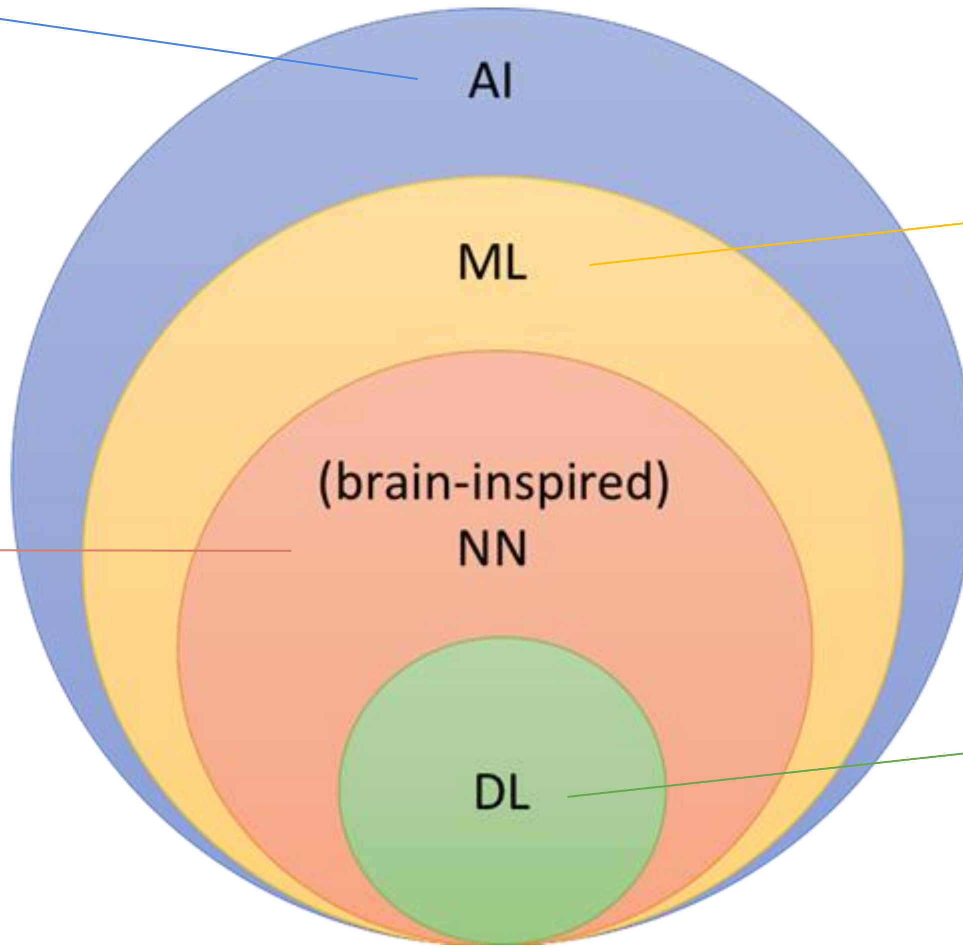
Where is AI heading?

- Artificial Intelligence (AI) is seamlessly integrating with our everyday life activities.
- Sometimes, we are <u>not even aware</u> of its use.

What is AI?

Any technique that enables a computer to mimic human behavior

Ability to learn without specifying a set of rules



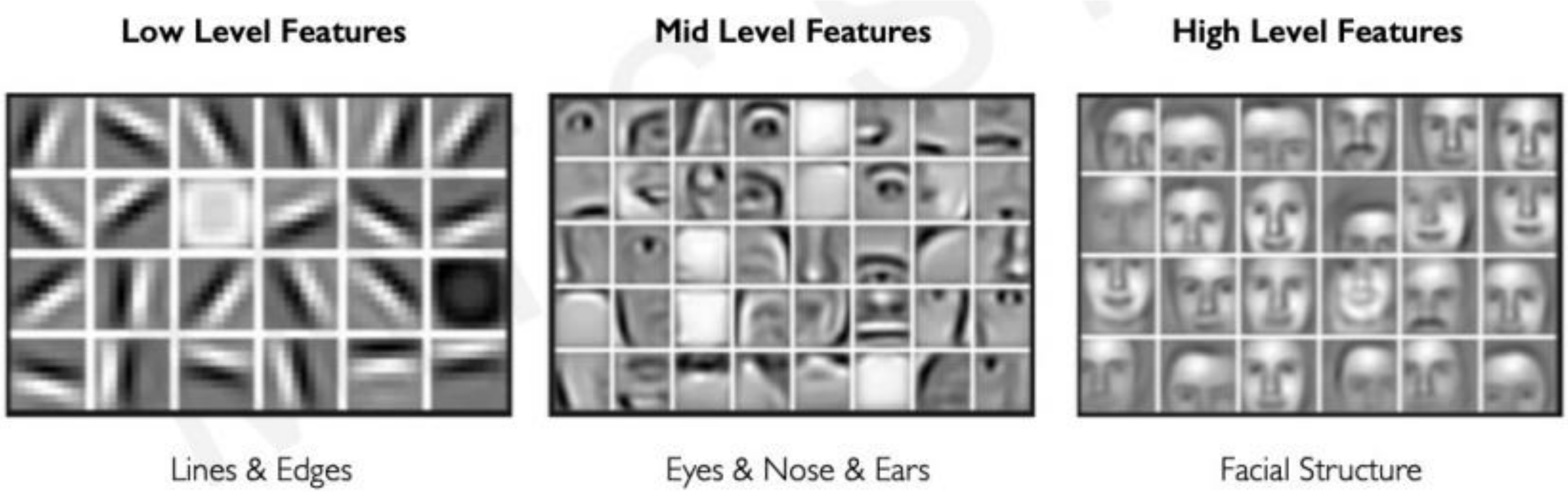
A subset of ML, inspired by the human brain

subset of ML using deep neural networks

Why Deep Learning?

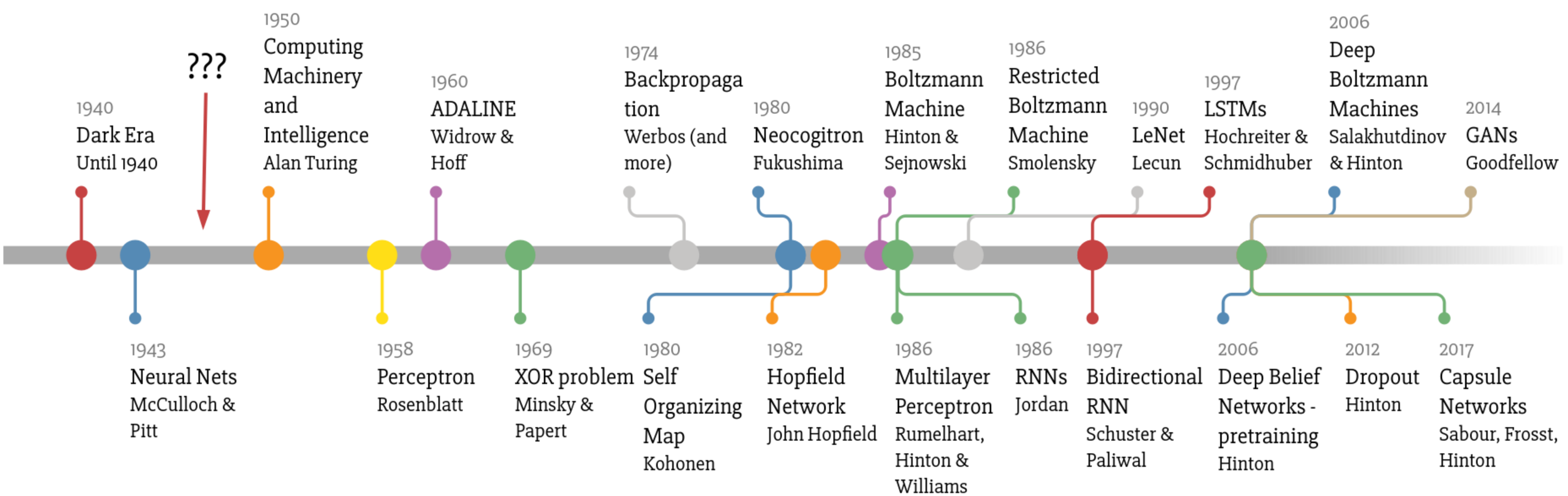
ML requires hand engineered features, which are expensive and time consuming to design.

Could we learn these features directly from the data?



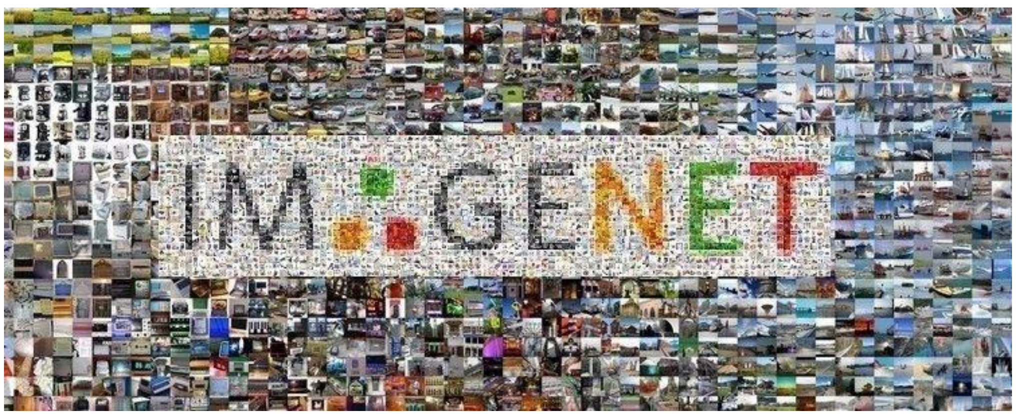
Why Now?

- Neural networks are not new. They date back decades ago.

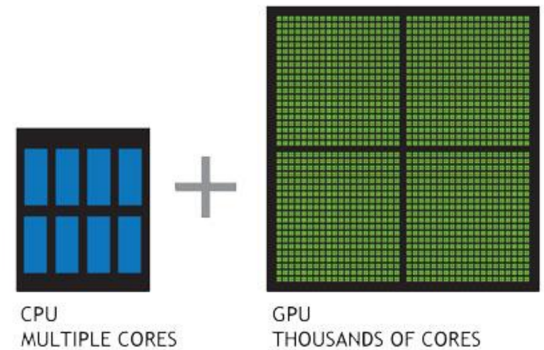


Why Now?

- so why the sudden dominance these days?



Abundance of data



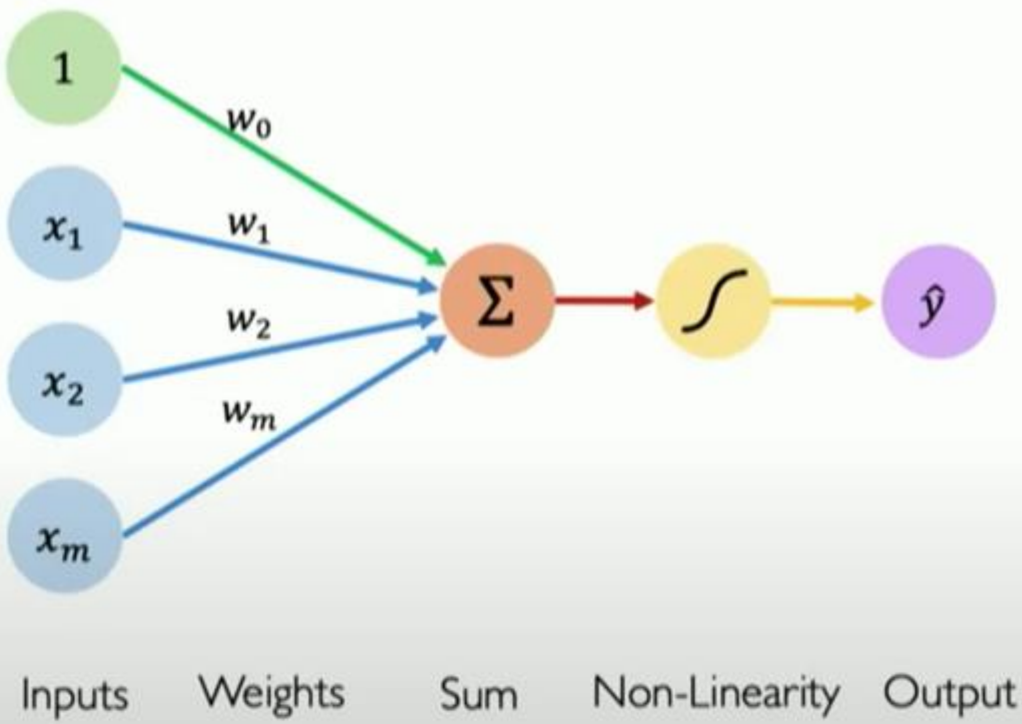
Advanced Hardware

TensorFlow

Advanced Programming Languages

How AI learns?

The Perceptron: Forward Propagation



Output

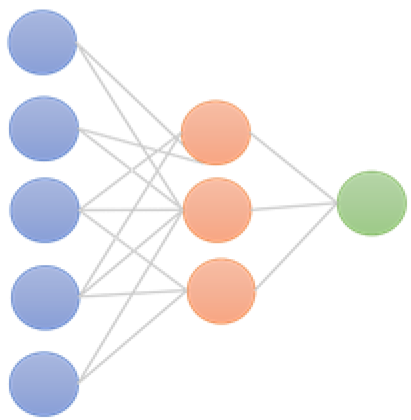
Linear combination of inputs

$$\hat{y} = g\left(w_0 + \sum_{i=1}^{m} x_i \, w_i \right)$$

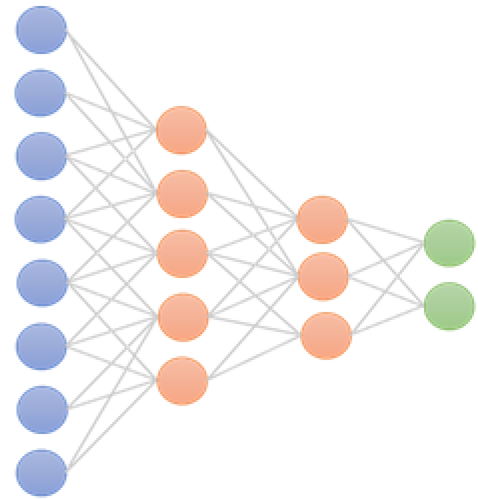
Non-linear activation function

Bias

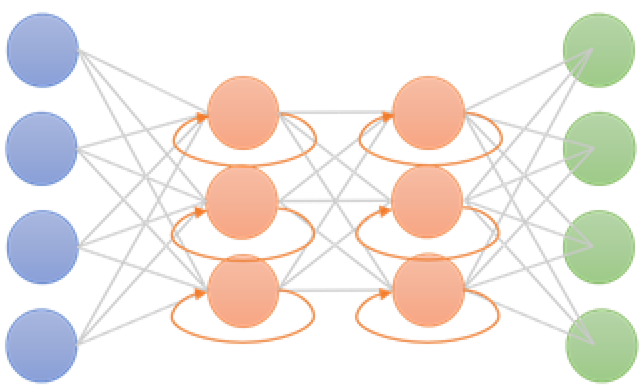
Types of NN



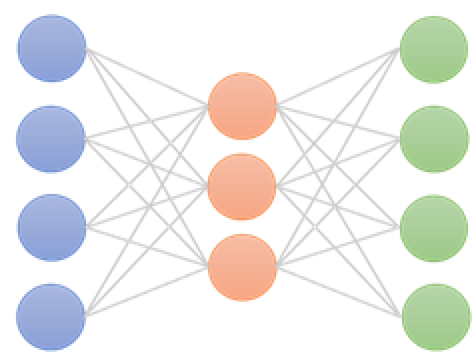
a) Feed-forward neural network



b) Convolutional Neural Network



c) Recurrent Neural Network



d) Autoencoders

Input Data

Data preprocessing is the process of:

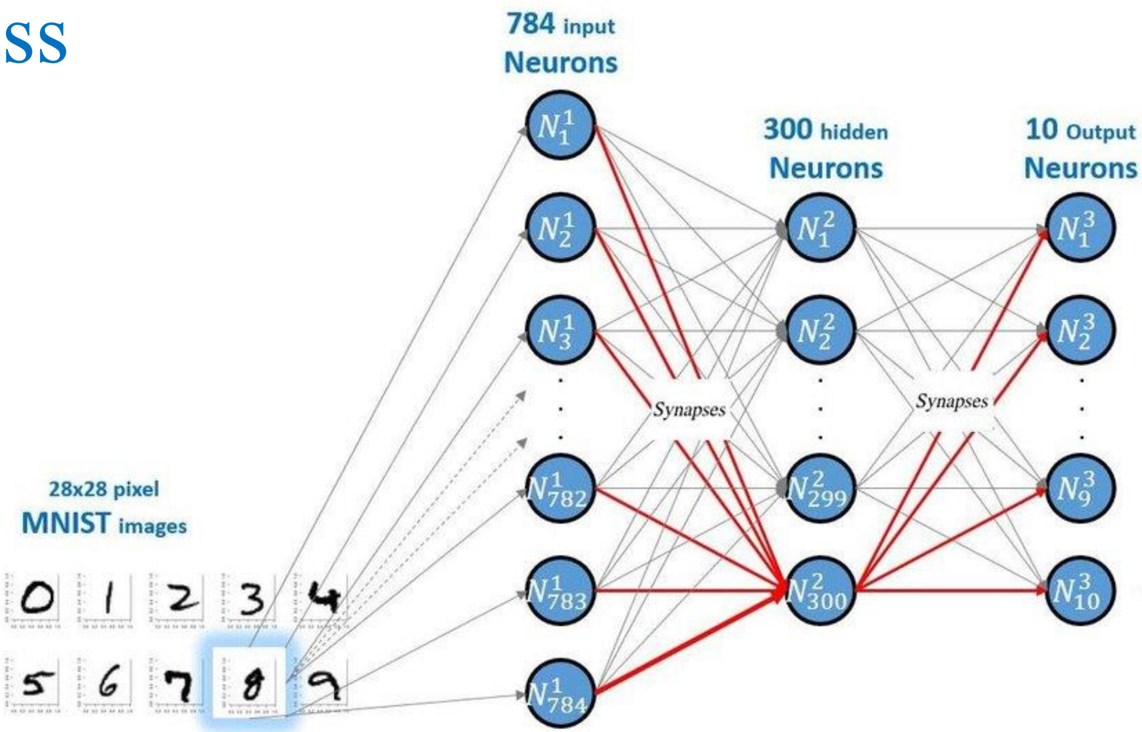
- evaluating,
- filtering,
- manipulating, and
- encoding data

The major goal of data preprocessing is to eliminate data issues:

- missing values,
- improve data quality,
- and make the data useful for machine learning purposes.

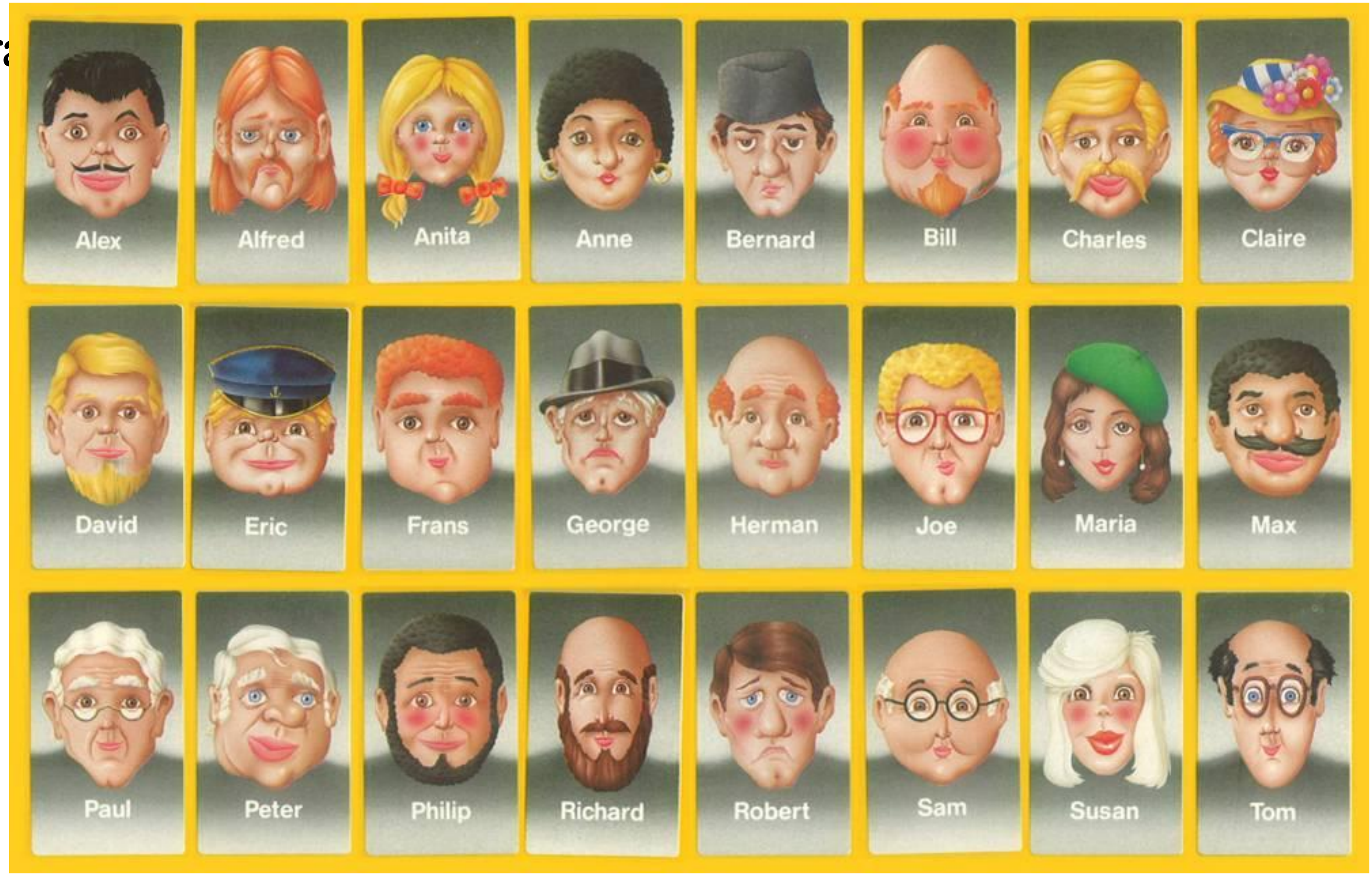
Training Neural Networks

The purpose of training neural nets is to find the optimal network weights that achieve the lowest loss



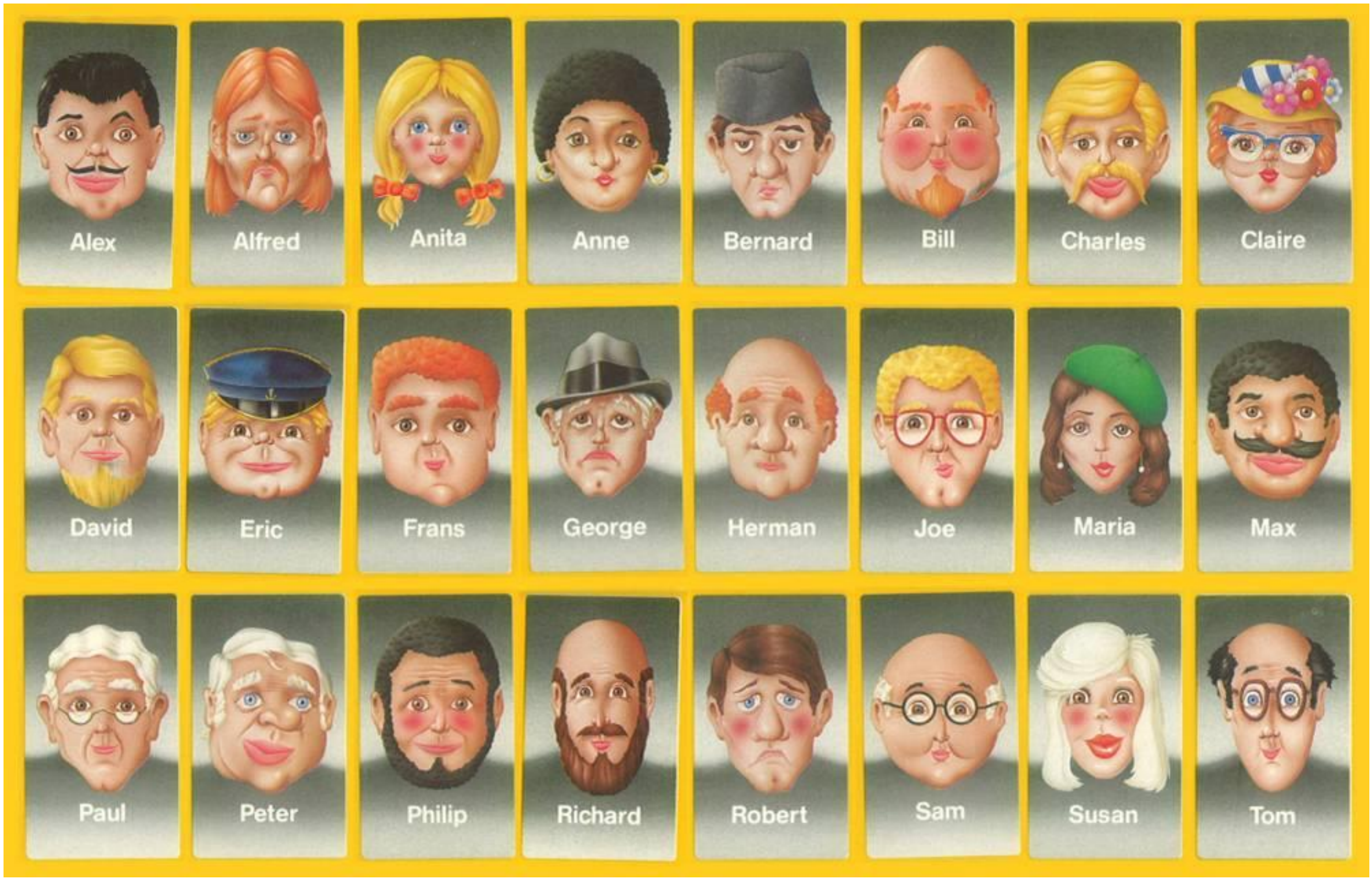
Loss Functions ?

Genera



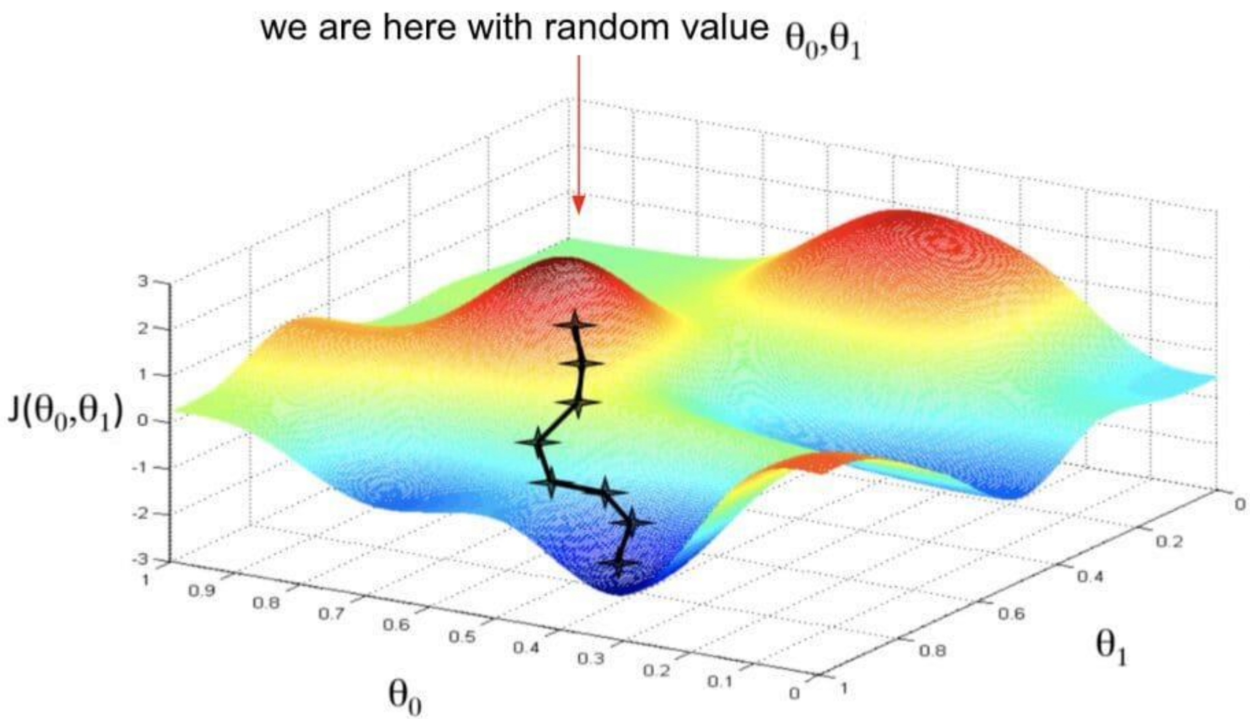
The lowest loss?

Player 1



Player 2

Loss function

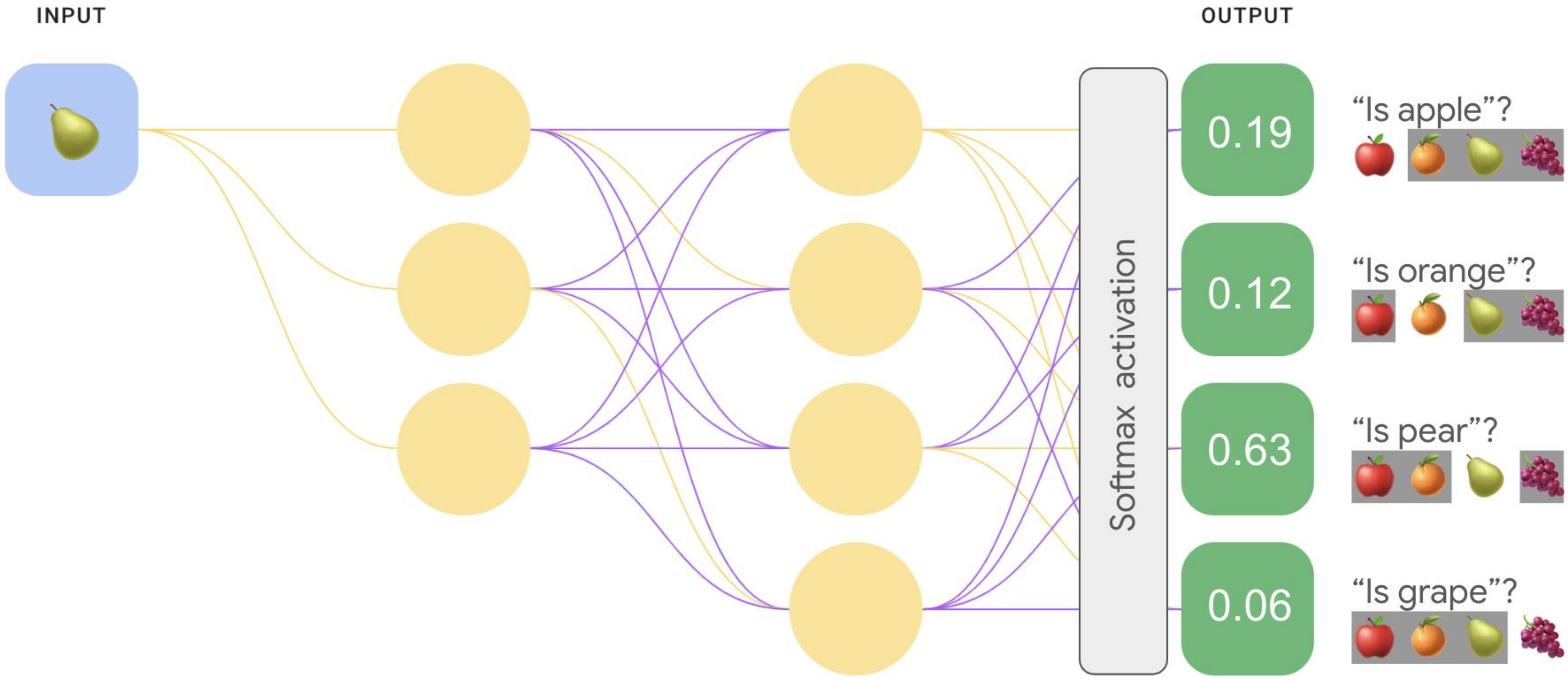


- Start with some θ_0, θ_1
- Keep changing θ_0, θ_1 to reduce $J(\theta_0, \theta_1)$

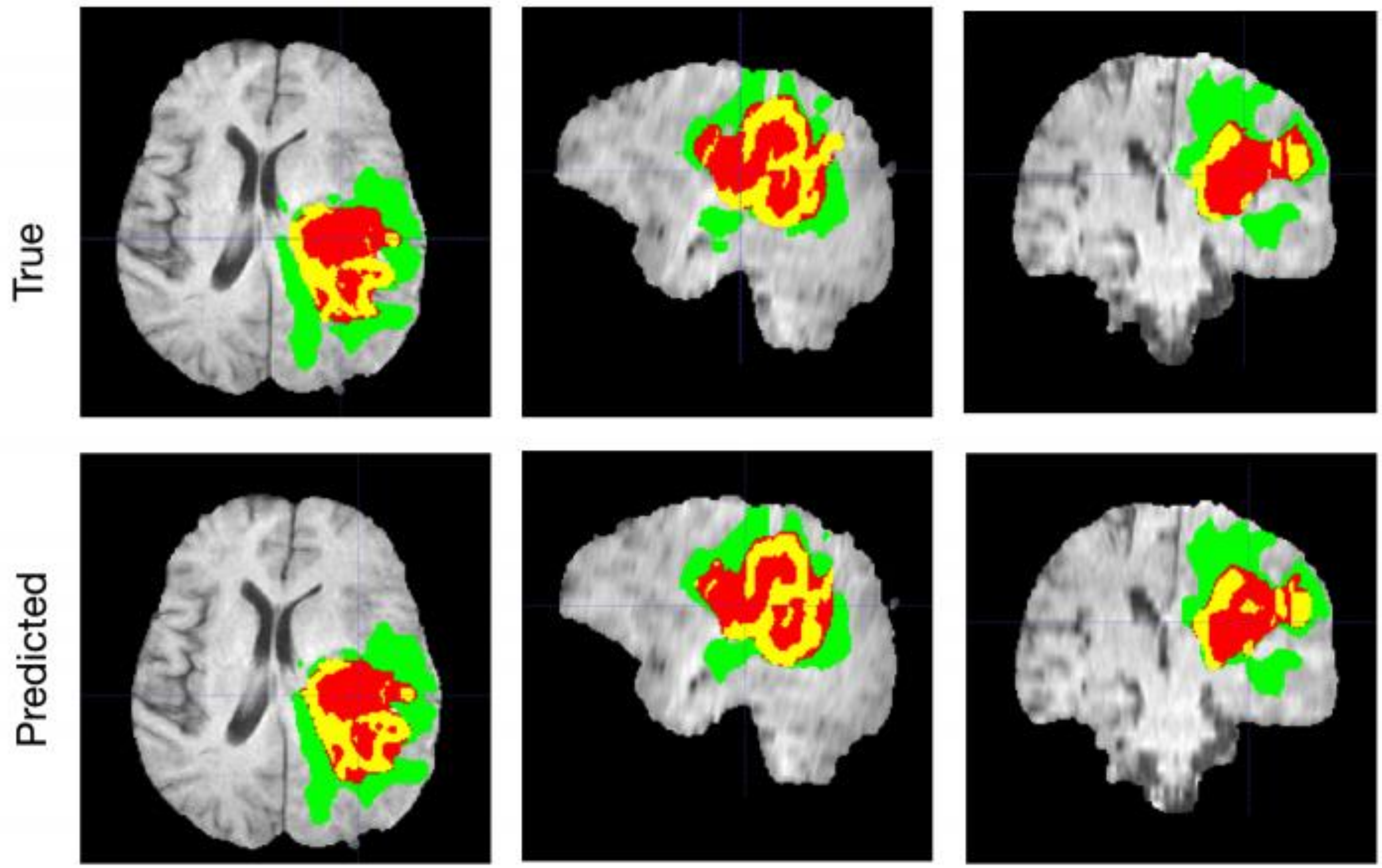
until we hopefully end up at a minimum

Gradient Descent (CALC I. Derivatives)

Output

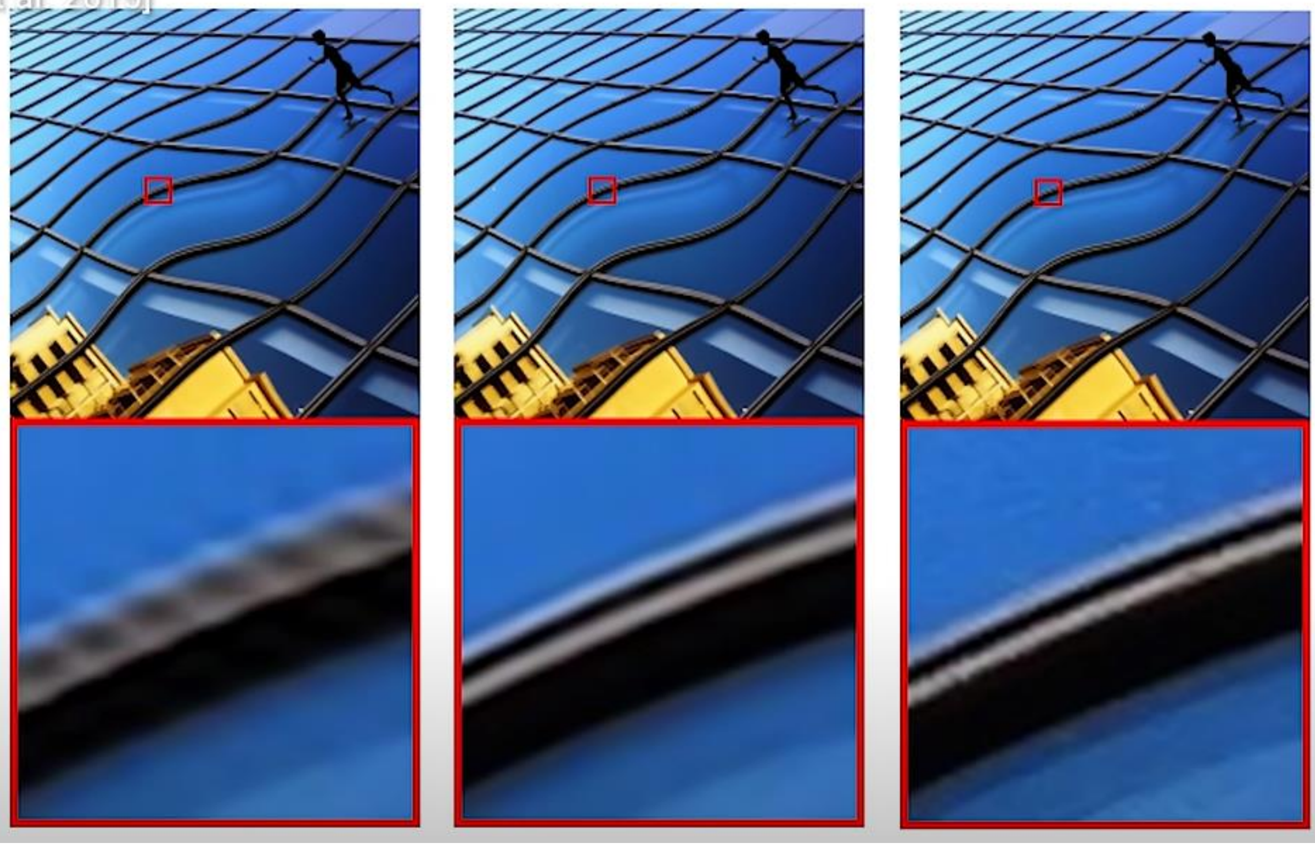

INPUT
OUTPUT
Softmax activation
0.19
0.12
0.63
0.06
“Is apple”?
“Is orange”?
“Is pear”?
“Is grape”?

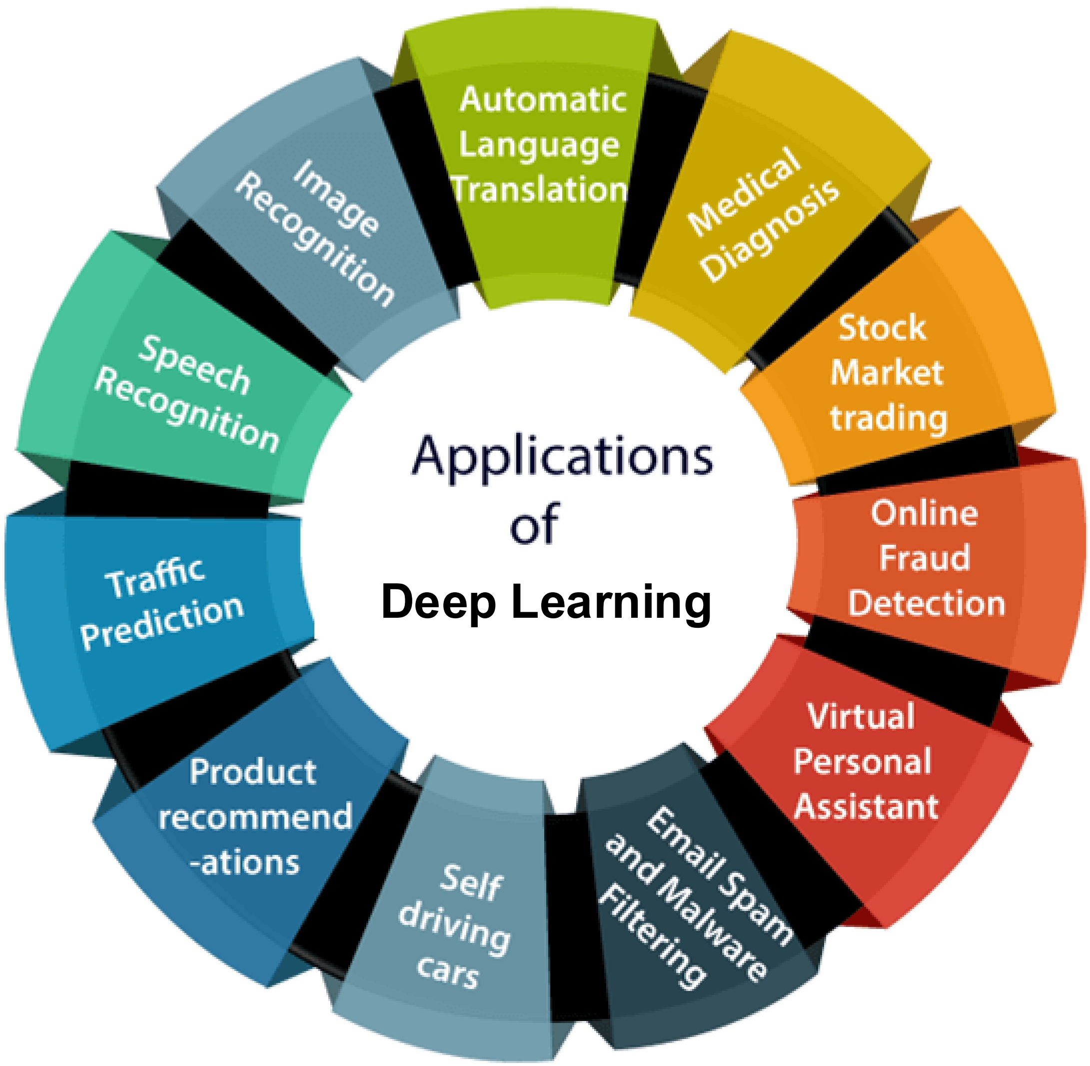
Brain Tumor Diagnosis



Other cool applications

Source: [Kim et al. 2015]




Applications
of
Deep Learning
Automatic Language Translation
Medical Diagnosis
Stock Market trading
Online Fraud Detection
Virtual Personal Assistant
Email Spam and Malware Filtering
Self driving cars
Product recommend -ations
Traffic Prediction
Speech Recognition
Image Recognition